**End-Term Examination**
**(CBCS)(SUBJECTIVE TYPE)(OffLine)**
**Course Name:<B.Tech IT/ CSE-AI/ CSE>, Semester:<5>**
**(November-December, 2023)**

| Subject Code: BCS 303 | Subject: Theory of Computation |
|---|---|
| Time :3 Hours | Maximum Marks :60 |

**Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.**

| | | | |
|---|---|---|---|
| Q1 | | (2.5*8=20) | |
| | (a) How many tuples are used to define a Finite Automata, explain with an example? | | |
| | (b) Design a DFA which accepts all the strings starting with 00 and ending with 101? | | |
| | (c) What is Greibach Normal Form, explain with an example? | | |
| | (d) Discuss about different closure properties of Context Free Grammar (CFG)? | | |
| | (e) What is the difference between Leftmost Derivation tree and Rightmost Derivation tree, draw the derivation tree for the string id+id*id using following productions of the grammar. E→E+E, E→E*E, E→(E), E→id | | |
| | (f) What is a Turing Machine Transducer? Design TM for 1's complement. | | |
| | (g) What is the difference between Context Free Language and Context Sensitive Language, explain with example? | | |
| | (h) What is halting problem, discuss about it briefly? | | |
| | UNIT-I | | |
| Q2 | (a) Find out whether the strings 00, 01001, 10010, 000, 0000 are accepted by the following Nondeterministic Finite Automata (NFA). | (10) | |

| | 0 | 1 |
|---|---|---|
| →$q_0$ | {$q_0$, $q_1$} | $q_1$ |
| *$q_1$ | $q_0$ | {$q_1$, $q_2$} |
| $q_2$ | ∅ | $q_1$ |

(b) Convert the above NFA to DFA, explain in detail.

| | | | |
|---|---|---|---|
| Q3 | Explain the steps followed in minimization of Finite Automata and minimize the following finite automata. | (10) | |

| | a | b |
|---|---|---|
| →$q_0$ | $q_1$ | $q_0$ |
| $q_1$ | $q_0$ | $q_2$ |
| $q_2$ | $q_3$ | $q_1$ |
| *$q_3$ | $q_3$ | $q_0$ |
| $q_4$ | $q_3$ | $q_5$ |
| $q_5$ | $q_6$ | $q_4$ |
| $q_6$ | $q_5$ | $q_6$ |
| $q_7$ | $q_6$ | $q_3$ |

| | | | |
|---|---|---|---|
| | UNIT-II | | |
| Q4 | How many tuples are used to define a Pushdown Automata (PDA) (describe each tuple) and design a PDA for L = $\{a^n b^{2n} : n>0\}$ on $\Sigma = \{a, b\}$. | (10) | |

**Q5** (a). Find out whether the grammar $S \rightarrow aSb/SS/\varepsilon$ is ambiguous of not. (Where $\varepsilon$ is epsilon – empty string). **(10)**

(b). Define Chomsky Normal Form (CNF). Convert the following Context free grammar to CNF.

$S \rightarrow bA/aB$

$A \rightarrow bAA/aS/a$

$B \rightarrow aBB/bS/b$

## UNIT-III

**Q6** By writing the Pumping Lemma for context free languages find out whether the language $L = \{a^n b^{2n} c^n / n>1\}$ is context free or not. **(10)**

**Q7** (a). Design a Turing Machine which accepts the Language $L = \{a^n b^n / n>1\}$ over $\Sigma = \{a,b\}$. **(10)**

(b). Discuss about Chomsky Hierarchy of Languages.

## UNIT-IV

**Q8** (a). Briefly describe about Complexity classes P and NP. **(10)**

(b). convert the following Moore Machine to Mealy Machine.

| Present State | Next state | | Output |
|---|---|---|---|
| | a=0 | a=1 | |
| $\rightarrow q_0$ | $q_3$ | $q_1$ | 0 |
| $q_1$ | $q_1$ | $q_2$ | 1 |
| $q_2$ | $q_2$ | $q_3$ | 0 |
| $q_3$ | $q_3$ | $q_0$ | 0 |

**Q9** Define Mealy Machine with tuples and convert the following Mealy Machine to Moore Machine. **(10)**

| Present state | Next state | | | |
|---|---|---|---|---|
| | Input a=0 | | Input a=1 | |
| | State | output | State | output |
| $\rightarrow q_1$ | $q_3$ | 0 | $q_2$ | 1 |
| $q_2$ | $q_1$ | 0 | $q_4$ | 0 |
| $q_3$ | $q_2$ | 1 | $q_1$ | 1 |
| $q_4$ | $q_4$ | 1 | $q_3$ | 0 |